
The Architecture of Errors: From Universal Impossibility to Patch-Local LLM Reliability

Anonymous Author(s)

Affiliation

Address

email

Abstract

Universal LLM reliability is not a finite-library problem: across all possible tasks, tools, schemas, knowledge sources, and evaluator expectations, new intervention-distinguishable failure modes can appear without bound, so no finite intervention dictionary can guarantee bounded residual error universally. But deployed systems do not operate over the whole universe; they operate inside bounded patches (legal review, medical RAG, code repair, customer-support agents, contract extraction) with recurring tasks, schemas, tools, and evaluator expectations. Within such patches, empirical evidence suggests failures are sparse, repetitive, and concentrated in a small recurring catalogue, so reliability becomes a local catalogue-discovery and intervention-coverage problem rather than an exponential token-length problem. We make this transition formal with two propositions: a negative result that no universal finite intervention dictionary exists, and a positive result that under logarithmic or capped mode discovery the patch-local intervention budget grows polylogarithmically in sequence length and becomes domain-constant once the patch catalogue saturates. The framework relocates rather than dissolves long-context difficulty: where the number of hard decisions itself grows with task length, reliability remains hard; the contribution is to identify the on-axis intervention rather than to make those regimes easy.

1 Introduction

The standard worry about long-context autoregressive generation is exponential. If every token has independent error probability e , then after n tokens the chance of a fully correct output is $(1 - e)^n$, which collapses to zero for any nontrivial e and large enough n [LeCun, 2023, Dziri et al., 2023]. Earlier work in this series [Anonymous, 2025] argued that this worry is misplaced because errors are not distributed uniformly across tokens: only 5–10% of tokens are “key” (genuinely dependent on long-range context [Fang et al., 2025]), while the remaining majority become nearly deterministic once enough context accumulates. Replacing $(1 - e)^n$ with the two-rate model $P(\text{correct}) = (1 - e_{\text{key}})^k (1 - e_{\text{non}})^{n-k}$ with $k \ll n$ scaling sublinearly in n recovers the observed long-context coherence and converts the reliability question from “how does n grow?” to “how does k grow?”.

From sparse tokens to recurring patterns. This paper takes the next step. Sparsity tells us *where* errors live; the natural follow-up is *what* they are. Recent failure-mode atlases supply a striking empirical answer: errors are not only sparse, they are also *repetitive*. ErrorAtlas [Ashury-Tahan et al., 2026], covering 83 models across 35 datasets, organises observed failures into 17 named categories sorted by prevalence, with a long-tailed distribution heavily concentrated in the head. In code, two

error types (AssertionError and NameError) cover 86.35% of failures on HumanEval across 14 LLMs, replicated across 23 models on HumanEval Pro and MBPP Pro [Wen et al., 2024]. In math, MWPEs-300K [Sun et al., 2025] categorises 304,865 errors from 15 LLMs across four math word-problem datasets and reports that dataset characteristics shape error patterns systematically. The same picture appears in multi-hop QA [Zhang et al., 2026], agentic tool use [Cemri et al., 2025], and retrieval-augmented generation [Wood and Forbes, 2024]. These observations suggest a third layer in the reliability architecture, complementing key-token sparsity (Layer 1) and within-key stratified-manifold structure (Layer 2 in Anonymous 2025). Within the small set of key tokens, only a fraction β produce *hard* failures, and inside bounded deployment patches those hard failures appear to cluster into a finite or effectively capped catalogue of recurring modes whose size grows much more slowly than the number of observed events.

Contributions. The central result is the conditional polylog intervention budget of Proposition 2: $m \geq \lceil |C|^{1-\varepsilon/e_{\text{hard}}} \rceil$, with a doubly-logarithmic rate as the optimistic special case under the postulate above. The *sequence*-level analogue is strictly tighter and approaches full-catalogue coverage as k grows; we surface this asymmetry rather than bury it. Around this central claim the paper does four supporting things. The three-layer formal framework, comprising sparsity (α), stratification (β), and a bounded mode catalogue ($|C|$), partitions key tokens into easy and hard subsets, then adds the recurring-pattern catalogue (§3). The empirical postulate of logarithmic mode discovery is calibrated against ErrorAtlas, HumanEval, and MWPEs, with $\sigma \in [0.87, 1.85]$ across anchors and $\sigma \approx 1.85$ adopted from ErrorAtlas as the conservative target. Section 4 synthesises evidence for clustering, cluster-selective interventions, and sublinear length scaling, drawing on ~ 60 prior published results including the six-axis capability-elimination harvest of 28 quantitatively-anchored citations stratified into Patterns A/B/C (Appendix B). And the re-audit of the most-cited steep-decay results (Appendix C) shows each decays over a variable other than raw token length.

2 Related Work

Failure-mode taxonomies. ErrorAtlas [Ashury-Tahan et al., 2026], MWPEs-300K [Sun et al., 2025], the HumanEval categorisation of Wen et al. [2024], and the RFMDataset [Guo et al., 2025] together suggest that, at any given corpus size, the number of distinct named failure modes is small (typically 8–20) and the cumulative coverage of the top modes is high. Domain-specific taxonomies for multi-hop QA [Zhang et al., 2026], multi-agent systems [Cemri et al., 2025], and tool-augmented agents [Yao et al., 2024] report the same pattern.

Targeted interventions. Each named cluster has been the subject of focused intervention research. Python execution closes most of the arithmetic-error cluster [Gao et al., 2023a, Chen et al., 2023, Gou et al., 2024b]; constrained decoding eliminates format violations [Suresh et al., 2025, Wang et al., 2025b, Dong et al., 2025]; execution feedback closes most code-logic errors [Shinn et al., 2023, Huang et al., 2023]; process reward models reduce step-level reasoning errors [Wang et al., 2024b, Lightman et al., 2024]; RAG strongly reduces hallucinations [Wood and Forbes, 2024]; preference optimisation closes over-refusal [Karaman et al., 2024]; structured uncertainty fixes most tool-call failures [Suri et al., 2025]. Two structural observations recur: each intervention is *cluster-selective* (residuals belong to a different named cluster, not to the targeted one), and *additivity is approximate but not perfect* [Patel et al., 2026, Le, 2026].

Length-decay benchmarks. A parallel literature measures the *shape* of the reliability decay curve. Mild-decay results (Loong [Wang et al., 2024a], GSM- ∞ [Zhou et al., 2025], RULER [Hsieh et al., 2024], anchor-based LLMs [Pang et al., 2024]) report log-linear, sigmoidal, or threshold-like decay inconsistent with smooth $(1 - \varepsilon)^n$. Steep-decay results [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026] are sometimes cited as exponential-compounding evidence. The apparent tension is reduced on careful reading: every steep-decay paper decays over a variable distinct from raw token length, which Appendix C documents.

Gap. What is missing is a quantitative bridge between the small, repeating taxonomy catalogue and a polylog intervention budget. We provide that bridge in §3.

85 3 Theoretical Framework

86 **Four levels of $|C|$.** Before any proposition, a definitional setup. LLM error analysis routinely
87 conflates four objects that the rest of this paper needs to keep apart:

- 88 • **L1: failure events.** Concrete LLM errors observed in a benchmark; the raw data behind any
89 taxonomy.
- 90 • **L2: empirical taxonomy categories.** Researcher-chosen labels grouping L1 events. ErrorAt-
91 las’s 17 categories, MWPE’s top-4 classes, and HumanEval’s `AssertionError/NameError`
92 partition all live here. L2 is observable but depends on the taxonomer’s resolution choices.
- 93 • **L3: latent failure modes.** The unobserved underlying clusters that L2 approximates. L3 is the
94 quantity Postulate 1 is morally about, but it cannot be measured directly; we treat L2 as a noisy
95 proxy for L3.
- 96 • **L4: capability axes / interventions.** The engineering unit: a Python interpreter, a constrained
97 decoder, a retrieval-augmented generator. L4 is coarser than L2 in practice; one capability axis
98 typically targets several L2 categories at once (§4, Claim B).

99 Throughout this section, $|C|$ refers to the **L2** count: that is what published taxonomies report.
100 Postulate 1 is therefore *engineering*-relevant because L4 is coarser than L2 (so a small intervention
101 library can sweep across many categories at once), and *epistemically* conditional because L2 is a noisy
102 proxy for the actual L3 mode catalogue whose faithfulness has not been independently measured.
103 Tagging the formal claims at the right level keeps the framework honest where casual error-talk slips.

104 **Roadmap.** The framework separates three questions that are often conflated. First, *where* do errors
105 occur? Only a sparse subset of key decisions. Second, *what* kinds of errors recur? A local catalogue
106 of failure modes. Third, *what fixes them?* A smaller library of capability interventions. The two
107 propositions of §3.4 formalise the transition from universal impossibility to patch-local tractability:
108 there is no finite intervention dictionary that covers all possible deployments, but inside a fixed
109 deployment patch the required library is small and slowly growing.

110 3.1 β -stratification of key tokens

111 The two-rate model of Anonymous [2025] distinguishes k key tokens (error rate e_{key}) from $n - k$
112 non-key tokens ($e_{\text{non}} \ll e_{\text{key}}$). Empirical atlases suggest that even within the key-token class errors
113 are not uniformly distributed: most key tokens are “decisions” for which the model has stable
114 representations; only a fraction concentrate the actual failures.

115 **Definition 1** (Hard fraction). Partition the k key tokens of a sequence into easy and hard subsets,

$$k_{\text{hard}} = \beta k, \quad k_{\text{easy}} = (1 - \beta)k, \quad \beta \in (0, 1),$$

116 where hard key tokens have an elevated error rate e_{hard} corresponding to manifold-transition decisions
117 in the sense of Anonymous [2025, §3.2], and easy key tokens have $e_{\text{easy}} \approx e_{\text{non}}$.

118 Under Definition 1, the composed sequence-level reliability becomes

$$P(\text{correct}) = (1 - e_{\text{hard}})^{\beta k} (1 - e_{\text{easy}})^{(1-\beta)k} (1 - e_{\text{non}})^{n-k}. \quad (1)$$

119 We do not assume iid token errors: the three rates summarise *conditional per-decision failure*
120 *hazards* after conditioning on prior decisions being correct, and grouping by stratum yields the
121 multiplicative survival expression. Because $e_{\text{easy}} \approx e_{\text{non}}$ once context accumulates, the failure rate
122 is dominated by the βk hard tokens, and we treat e_{hard} as the load-bearing quantity. The parameter
123 β is *latent*: failure-mode atlases report $\Pr(\text{category} \mid \text{error occurred})$, which does not identify
124 $\Pr(\text{hard} \mid \text{key-token decision})$. We carry β symbolically.

125 3.2 Empirical postulate: logarithmic mode discovery

126 Stratification names a target, the βk hard-token decisions, but does not yet tell us how many distinct
127 ways those hard decisions can fail. The empirical answer from Section 1 is that failures repeat: a
128 small number of recurring patterns covers most of the mass. The next question is how the catalogue’s

size grows with the number of observed failures, because that is the quantity an intervention library has to keep up with.

Two candidates exist in the type-token literature: Heaps’ law (power-law, $|C| \approx K \cdot k_{\text{hard}}^b$ with $b \in [0.4, 0.6]$ [Manning et al., 2008]), and logarithmic discovery. **Zipfian rank-frequency does not imply logarithmic growth.** Heaps’ law is the correct type-token consequence of Zipf-distributed events, and it is power-law, not logarithmic. We therefore state logarithmic mode discovery as an empirical postulate, defensible by direct measurement of error taxonomies, not as a theorem.

Two sample-size variables matter, and they are easy to conflate. Let $h(n) = \beta k(n)$ denote the number of hard-token decisions in a single sequence of length n ; let T denote the number of observed hard-failure events in a corpus used to discover the domain catalogue. These are not the same object: $h(n)$ controls per-sequence exposure, T controls empirical discovery. We write C_D for the full reachable catalogue inside domain patch D , $C_{\text{seen},D}(T)$ for the subset discovered after T sampled failures, and $C_{\text{active},D}(n)$ for the subset a single sequence of length n can activate.

Postulate 1 (Patch-indexed catalogue discovery). Within a fixed application domain D , the number of named recurring failure modes discovered after T sampled hard-failure events is bounded by

$$|C_{\text{seen},D}(T)| \leq \min(A_D + \sigma_D \ln T, |C_D|), \quad \sigma_D > 0, A_D \geq 0.$$

The cap $|C_D|$ is the domain-imposed ceiling. The postulate concerns catalogue *discovery*, not the number of hard decisions inside a single sequence.

Assumption 2 (Per-sequence active-mode exposure). For a single sequence of length n in domain D ,

$$|C_{\text{active},D}(n)| \leq \min(A'_D + \sigma'_D \ln h(n), |C_D|).$$

Primed constants are distinct from those of Postulate 1: corpus discovery and per-sequence activation need not share the same rate.

The two bounds answer different questions, and the rest of the paper picks the right one at each point. Proposition 2’s sequence-length scaling rate is governed by $C_{\text{active},D}(n)$ via Assumption 2: how much of the catalogue can a single sequence touch? Full domain-library budgeting, the practical engineering target for a deployed system, uses C_D : how large does the library need to be to cover the domain’s reachable failure modes? The cap $|C_D|$ enters both: once enough has been sampled (or activated) to saturate the domain ceiling, further growth stops and the budget becomes a domain-constant function of $|C_D|$ alone.

Empirical defense. Three published taxonomies anchor the mode-rate parameter σ at endpoint corpus scales. ErrorAtlas at $T \gtrsim 10^4$ observed failures has $|C| = 17$, giving $\sigma \approx 17 / \ln T \approx 1.85$ at $T = 10^4$ under $A = 0$. HumanEval [Wen et al., 2024], on a comparable corpus size, has $|C| \in [8, 12]$ and accordingly $\sigma \in [0.87, 1.30]$. MWPEs, an order of magnitude larger ($T \approx 3 \times 10^5$), still has $|C| \in [15, 20]$ and yields $\sigma \in [1.2, 1.6]$. Across the three anchors, σ falls in the range $[0.87, 1.85]$; we carry $\sigma \approx 1.85$ (the ErrorAtlas value, the largest of the three) as a conservative downstream target. Endpoint counts at a single corpus scale are not discovery curves: a subsample-vs-discovered-modes measurement has not been published for any LLM error taxonomy at this writing, and remains the explicit empirical test we flag (§6, Appendix A).

Capabilities are coarser than error classes. Postulate 1’s $\sigma \approx 1.85$ is calibrated against *named error categories*, but a single capability axis often targets several L2 categories at once. A Python interpreter removes the execution-error component of arithmetic, unit conversion, simple counting, list manipulation, and date arithmetic; a constrained decoder eliminates by construction the format-violation cluster and the structural component of “missing required element.” Postulate 1 is therefore a conservative input for the engineering question; the formal coverage problem is weighted set cover, with the ranked-category form of Proposition 2 as a tractable proxy. An alternative power-law (Heaps) formulation and symbolic-form sensitivity analysis across candidate cluster-count laws are given in Appendix A.

Domain patches cap the catalogue. A model deployed in a fixed application domain occupies a bounded sub-region of the stratified manifold, a *domain patch* P_D [Anonymous, 2025, Li and Sarwate, 2025]. Empirical support: embedding-space intrinsic dimensions of ~ 5 –15 across domains, well below ambient $\sim 10^3$ [Park et al., 2024, Li and Sarwate, 2025]; cross-domain accuracy spreads of 20–70% on a single base model [Wang et al., 2024c]; log-popularity thresholds below which

scaling fails [Mallen et al., 2023, Kandpal et al., 2023]. None of these *measure* $|C_D|$ directly; together they motivate the patch-indexed hypothesis. The patch makes logarithmic or saturating discovery dynamics plausible inside fixed domains; σ_D , A_D , β_D , $|C_D|$ are all domain-indexed. We treat the patch’s reachable catalogue as finite or effectively capped: a modelling choice motivated by the heterogeneity evidence above, not a derivation from boundedness alone.

3.3 Coverage by a targeted intervention library

With the catalogue defined, the next question is how much residual error a library of m targeted interventions actually closes off. Given a catalogue of $|C|$ recurring modes, let $p_1 \geq p_2 \geq \dots \geq p_{|C|}$ denote the ranked empirical hard-error mass of those modes with $\sum_i p_i = 1$, so a library covering the top m modes removes cumulative error mass $\sum_{i=1}^m p_i$. We adopt the log-coverage form

$$F(m; |C|) = \min\left(1, \frac{\ln m}{\ln |C|}\right) \quad (2)$$

as a *postulated* parametric approximation to that cumulative top- m mass, not as a raw count of categories covered. Reading the formula: the first few interventions cover disproportionate mass (the head of the distribution is dense), then incremental coverage falls off as the library saturates the catalogue. For our ErrorAtlas anchor ($|C| = 17$, $m = 5$), $F(5; 17) \approx 56.8\%$; doubling the library to $m = 10$ pushes it to $F(10; 17) \approx 81.3\%$.

This is an empirical best-fit choice, not derived from Zipf. ErrorAtlas sorts its 17 categories by prevalence but does not publish a single cross-domain cumulative top- m curve at this writing; alternative cumulatives (Zipf–Mandelbrot, truncated power law, saturating exponential) match the same qualitative shape with comparable freedom. Appendix A shows the polylog conclusion of §3.4 survives across the family.

Two caveats matter for what comes next. (a) F tracks cumulative error mass across ranked L2 categories, not L4 capability axes deployed; in real deployments, where one capability often sweeps several L2 categories, residual error at a given m is generally lower than (2) predicts. (b) The form is intended for $m \geq 2$, $|C| \geq 2$; we adopt $F(0; |C|) := 0$ and $F(1; |C|) := p_1$ (the mass of the single most prevalent mode) since the log expression is not anchored at $m = 1$. After deploying the top- m library, the residual per-hard-token error rate is

$$e_{\text{res}}(m) = (1 - F(m; |C|)) e_{\text{hard}}. \quad (3)$$

3.4 From universal impossibility to patch-local reliability

The finite-catalogue claim is not universal. Before deriving the local intervention budget, we first state the corresponding negative result: if a domain keeps generating intervention-distinguishable failure modes, no finite dictionary can cover it. The positive result of this paper begins only after a deployment patch D has been fixed.

Proposition 1 (No Universal Finite Intervention Dictionary). *Fix a residual-error tolerance ε . Say that an intervention covers a failure mode if it reduces that mode’s residual error below ε . A domain D is intervention-unbounded if it contains an infinite sequence of failures $(f_j)_{j=1}^\infty$ such that each f_j is not covered by any finite intervention dictionary that covers $\{f_1, \dots, f_{j-1}\}$. Then the ε -resolution failure catalogue C_D^ε is infinite. Consequently, no finite intervention dictionary can guarantee residual error below ε over all of D .*

Proof. Suppose C_D^ε were finite. Then after all its modes had appeared, every later failure would belong to one of the already-seen modes, so the interventions covering the earlier representative of that mode would also cover the later failure. But intervention-unboundedness says that each new f_j is not covered by any dictionary covering the earlier failures, contradicting the assumption. Therefore C_D^ε is infinite. Now suppose for contradiction that some finite dictionary $\mathcal{I}$ covered all of D . Then $\mathcal{I}$ in particular covers $\{f_1, \dots, f_{j-1}\}$ for every j , so by intervention-unboundedness $\mathcal{I}$ does not cover f_j ; but $\mathcal{I}$ covers all of D , so it covers f_j . Contradiction. $\square$

Engineering interpretation. A universal reliability library is not a well-posed target. If the domain keeps changing the task family, tools, schemas, knowledge sources, and evaluator expectations, then

new intervention-distinguishable failures can appear indefinitely. The correct reliability object is therefore not “all LLM behaviour,” but a bounded deployment patch. Proposition 1 is the inoculation against a misread of the rest of this paper: we are not claiming a fixed list of ≈ 50 named patterns covers LLM use in general. We are claiming that *inside a fixed operational patch* the reachable catalogue can be modelled as finite or effectively capped, and that is the regime in which the engineering math below applies.

From the negative to the positive result. Proposition 1 explains why universal long-context reliability is the wrong engineering object. The right object is a bounded deployment patch. Once D is fixed, the question changes from “can one library cover all possible LLM failures?” to “how large must the library be to cover the reachable catalogue C_D inside this patch?” We now compose the three ingredients of §§3.2–3.3, namely β -stratification (Definition 1), the active-mode bound (Assumption 2), and the log-coverage form (Eq. 2), into that local budget statement. For *sequence-length scaling* we use Assumption 2, not Postulate 1: the empirical sparsity $\alpha = k/n \in [0.05, 0.10]$ is a *finite-length* statement, and under Anonymous [2025, §3.1]’s $k \sim \log n$ regime, α decays as $O(\log n/n)$.

Conditionality. The bound below is engineering math, not a theorem about LLMs. It holds conditional on three things: (i) the log-coverage form of §3.3 applied on its declared domain $m \geq 2$, $|C_{\text{eff}}| \geq 2$; (ii) a residual target $\varepsilon \in (0, e_{\text{hard}})$ that is non-trivial but attainable; (iii) Assumption 2 for any sequence-length rate claim, with the patch hypothesis that a bounded operational patch justifies modelling the reachable catalogue as finite or effectively capped. A different cumulative fitted to the same ErrorAtlas/HumanEval/MWPES anchors would preserve the qualitative polylog conclusion (Appendix A) while changing the exponent.

Proposition 2 (Patch-Local Intervention Budget). *Let $\varepsilon \in (0, e_{\text{hard}})$ be a target per-hard-token residual error rate for sequences in domain D , with $|C_{\text{eff}}| \geq 2$, where*

$$C_{\text{eff}} = \begin{cases} C_{\text{active},D}(n) & (\text{per-sequence scaling}), \\ C_D & (\text{full domain-library deployment}). \end{cases}$$

Under the coverage form of §3.3, the smallest intervention library satisfying $e_{\text{res}}(m) \leq \varepsilon$ has size

$$m \geq \left\lceil |C_{\text{eff}}|^{1-\varepsilon/e_{\text{hard}}} \right\rceil. \quad (4)$$

If $C_{\text{eff}} = C_{\text{active},D}(n)$ and $k(n) = \Theta(\log n)$, then in the pre-cap regime

$$m = O((\log \log n)^{1-\varepsilon/e_{\text{hard}}}),$$

i.e., the intervention budget grows doubly logarithmically with sequence length. In the cap regime, or for full domain-library deployment, $m \geq \lceil |C_D|^{1-\varepsilon/e_{\text{hard}}} \rceil$ independent of n .

Proof. From §3.3, $e_{\text{res}}(m) = (1 - F(m; |C_{\text{eff}}|)) e_{\text{hard}}$. Requiring $e_{\text{res}}(m) \leq \varepsilon$ gives $F(m; |C_{\text{eff}}|) \geq 1 - \varepsilon/e_{\text{hard}}$. Since $\varepsilon < e_{\text{hard}}$, the required coverage lies below one and the cap in F is inactive; with $F = \ln m / \ln |C_{\text{eff}}|$, we obtain $m \geq |C_{\text{eff}}|^{1-\varepsilon/e_{\text{hard}}}$, and taking the ceiling gives (4). Assumption 2 gives $|C_{\text{active},D}(n)| = O(\log \log n)$ when $k(n) = \Theta(\log n)$, before the cap. After the cap, $|C_{\text{eff}}| = |C_D|$ and the bound is independent of n . $\square$

Engineering interpretation. Once the patch is fixed, the problem changes. The relevant question is no longer whether arbitrary future failures exist, but how quickly the reachable local catalogue is discovered and how much residual mass is removed by the top interventions. Under the paper’s assumptions, this yields a small and slowly growing intervention budget for per-hard-token reliability, domain-constant in the cap regime, with the exact size determined by the local discovery and rank-coverage curves rather than by a universal prior.

We label these results *Propositions* rather than *Theorems* to keep their conditional nature visible. The formal content is engineering math, derived from the empirical postulates of §3.2 and §3.3; it formalises a directional intuition rather than deriving a law of LLM reliability. The doubly-logarithmic rate stated in (4) is the *optimistic special case*: per-sequence, pre-cap, under logarithmic mode discovery. Weaker (but still polylog) rates obtain under Heaps (Appendix A); the cap regime is a domain-constant independent of n .

270 **Sequence-level versus per-hard-token bounds.** Proposition 2 bounds the *per-hard-token* residual
 271 error. Sequence-level failure probability is strictly stronger. Define the non-hard-token survival factor

$$S_{\text{base}} = (1 - e_{\text{easy}})^{(1-\beta)k} (1 - e_{\text{non}})^{n-k}, \quad (5)$$

272 so that the composed survival from (1) can be written $P(\text{correct}) = (1 - e_{\text{res}})^{\beta k} S_{\text{base}}$. The sequence-
 273 level target $P(\text{correct}) \geq 1 - \varepsilon_{\text{seq}}$ then yields three regimes:

274 **Regime (i):** $S_{\text{base}} < 1 - \varepsilon_{\text{seq}}$. Even at $e_{\text{res}} = 0$ the surviving non-hard-token failure mass exceeds
 275 the sequence-level budget. Hard-token interventions alone cannot meet the target; this is the
 276 route by which the exponential-in- n concern re-enters, and it should be diagnosed before any
 277 catalogue-budgeting exercise.

278 **Regime (ii):** $S_{\text{base}} \geq 1 - \varepsilon_{\text{seq}}$. Define

$$\tau_{\text{seq}} = 1 - \left(\frac{1 - \varepsilon_{\text{seq}}}{S_{\text{base}}} \right)^{1/(\beta k)}. \quad (6)$$

279 For $0 < \tau_{\text{seq}} < e_{\text{hard}}$, the sequence target is met whenever $e_{\text{res}}(m) \leq \tau_{\text{seq}}$, so the required library
 280 size is

$$m \geq \left\lceil |C_{\text{eff}}|^{1-\tau_{\text{seq}}/e_{\text{hard}}} \right\rceil.$$

281 As the target tightens toward the feasibility boundary, $\tau_{\text{seq}} \rightarrow 0$ and $m \rightarrow |C_{\text{eff}}|$, so essentially
 282 every named mode must be covered.

283 **Regime (iii):** $\tau_{\text{seq}} \geq e_{\text{hard}}$. The sequence target is already met at the baseline hard-token rate; no
 284 intervention is required.

285 A common engineering rule of thumb, “fix ≈ 50 patterns per domain $\rightarrow \approx 90\%$ error reduction,”
 286 is best understood as a per-hard-token statement; the analogous sequence-level claim requires
 287 correspondingly more interventions and tighter coverage of the tail. Production systems with one-shot
 288 reliability targets should plan for the stricter regime, and check whether they are in Regime (i), where
 289 catalogue-budgeting cannot help at all.

290 4 Empirical Evidence

291 We collect evidence for three load-bearing claims: errors cluster into a small recurring set (A); each
 292 cluster is addressable by one targeted intervention (B); reliability decays sublinearly with output
 293 length (C).

294 **Claim A: Failure-mode clustering.** The strongest single result is ErrorAtlas [Ashury-Tahan
 295 et al., 2026]: 83 models $\times$ 35 datasets, $\approx 7,000$ failures, 17 named categories with a long-tailed,
 296 head-concentrated prevalence ordering. Domain-specific Paretos reproduce the shape inside each
 297 domain: math (MWPEs top-4 categories dominate per model with strong cross-model overlap [Sun
 298 et al., 2025]), code (AssertionError+NameError = 86.35% on HumanEval, replicated across 23
 299 models [Wen et al., 2024]), multi-hop QA (a single dominant “missing-evidence” mode [Zhang
 300 et al., 2026]), agentic tool use (under 20 recurring modes across two studies [Cemri et al., 2025,
 301 Yao et al., 2024]), and RAG (a handful of distinguishable hallucination types [Wood and Forbes,
 302 2024]). The catalogue is also *stable* across models: ErrorAtlas reports a fixed category hierarchy
 303 across 83 models; RFMDataset [Guo et al., 2025] finds “strikingly similar failure mode distributions”
 304 across ten advanced reasoning models; EDIT [Dai et al., 2025] measures a $\approx 4.7\%$ key-step fraction
 305 stable across models. Schaeffer et al. [2025] provide the closest theoretical anchor, proving that the
 306 observed aggregate power-law scaling of LLM evaluations requires the per-problem success-rate
 307 distribution to be heavy-tailed near $p = 0$, structurally related to Postulate 1.

308 **Claim B: Cluster-selective capability interventions.** A dedicated harvest yields 28 capability-
 309 elimination citations across six independent axes (arithmetic, code execution, format/structure,
 310 perception/grounding, knowledge/RAG, verification), each axis independently confirmed by between
 311 three and nine citations. The full table and the three structural patterns (Pattern A by-construction, B
 312 strong-empirical-with-class-shift, C moderate-with-shift) appear in Appendix B.

Arithmetic is the cleanest case: PAL [Gao et al., 2023a] lifts GSM-Hard from 20.1% to 61.5%, and the residuals are problem-comprehension errors rather than arithmetic ones. Format violations admit something stronger still: constrained decoding zeroes invalid-token probability by construction [Suresh et al., 2025, Wang et al., 2025b, Dong et al., 2025]. Code-logic errors yield to execution feedback (Reflexion + AgentCoder push HumanEval pass@1 from 80% to 96.3%, with residuals in spec-misinterpretation [Shinn et al., 2023, Huang et al., 2023]); reasoning steps to process supervision (Math-Shepherd: Mistral-7B MATH 28.6% $\rightarrow$ 43.5% [Wang et al., 2024b]); hallucinations to RAG, where Acurai [Wood and Forbes, 2024] reports 100% elimination on RAGTruth, a benchmark-conditional empirical result rather than a general by-construction guarantee. Two further clusters complete the picture: POROver lifts the not-overrefused rate from 57.6% to 82.1% [Karaman et al., 2024], SAGE-Agent raises When2Call accuracy from 36.5% to 65.2% [Suri et al., 2025].

The class-shift signature, in which post-intervention residuals belong to structurally different classes, is the empirical content of the cluster-orthogonality claim that underwrites Proposition 2’s composition step. Capabilities are coarser than error classes: a single Python interpreter removes execution-error components from five named clusters; constrained decoding eliminates format and the structural component of “missing required element” jointly. A negative control sharpens the selectivity claim. DebugBench [Tian et al., 2024] reports that execution feedback works for syntax and reference errors and is explicitly “unhelpful for logic errors,” exactly where the framework predicts capability provisioning fails.

Claim C: Sublinear length scaling. Direct evidence for sublinear (rather than exponential) decay shows up across very different measurement designs. Loong [Wang et al., 2024a] reports GPT-4o on Chain-of-Reasoning declining 81.6% $\rightarrow$ 32.9% across the 10K $\rightarrow$ >200K context bins; log-linear over $\log L$ and not well explained by smooth independent per-token exponential decay. GSM- ∞ [Zhou et al., 2025] attacks the independent-error model from a different angle: exponentially more inference compute yields only *linear* AUC gains, and DeepSeek-R1 maintains 10%+ accuracy at 130 reasoning operations where $(1 - \varepsilon)^{130}$ for any reasonable ε predicts near-zero. Press et al. [2023] find a $\approx 40\%$ compositionality gap that is approximately *constant* across the GPT-3 family, direct evidence against per-step independent compounding (which would predict the gap to grow with chain length).

Structural correlates of the same phenomenon appear elsewhere. Pang et al. [2024]’s Anchor LLMs achieve $\approx 99\%$ K/V reduction with only minor accuracy compromise: the effective number of distinct attention-key vectors a model needs is far smaller than n . Think-Prune-Train [Costello et al., 2025] raises Pass@1 while Pass@20 plateaus, the manifold-transition signature. RULER [Hsieh et al., 2024] finds threshold behaviour inconsistent with smooth $(1 - \varepsilon)^n$. METR [Kwa et al., 2025] fits logistic-in-log-length, which is mathematically sublinear in length itself.

Self-consistency [Wang et al., 2023] is consistent with but does not *prove* clustered structure: Condorcet aggregation of iid Bernoulli chains can yield similar gain magnitudes, so we treat it as suggestive rather than diagnostic. Prominent steep-decay counter-evidence [Dziri et al., 2023, Kuratov et al., 2024, Wan et al., 2026, Karpinska et al., 2024] decays over variables distinct from raw token length; per-paper re-audits showing the relocation pattern are in Appendix C.

5 Practical Implications

Reliability engineering is local patch coverage. Within a fixed domain patch, Proposition 2 makes reliability a small-catalogue engineering problem rather than an asymptotic scaling problem. A team should initially budget for a library on the order of tens of interventions, then refine from the local mode-discovery curve $C_{\text{seen}, D}(T)$ and the empirical rank-frequency distribution. As a rule of thumb consistent with the head-mass figures in ErrorAtlas, HumanEval, and MWPEs, ≈ 50 named interventions cover the bulk of the per-hard-token failure distribution in many measured domains. This is a planning prior calibrated to current taxonomies, not a universal constant; the local mode-discovery curve sets the actual library size for any given deployment. The same base model in cardiology RAG, legal contract drafting, and code review yields three different libraries because C_D , A_D , σ_D all change with the deployment domain.

Per-hard-token vs. sequence-level targets matter for SLA design. Per-hard-token residual error, the right metric for continuous-correction systems, has a polylog budget. Sequence-level failure

probability, the right metric for one-shot systems, is strictly stricter and approaches full-catalogue coverage as k grows. Production deployments should design SLAs to match the actual cost structure, not read the per-token result as the production SLA.

Library design as engineering, not asymptotic theory. An intervention library can be assembled module-by-module, in any order, as long as modules target distinct failure classes. Layer-separated interventions (constrained decoding + retrieval + process supervision + tool call) compose approximately additively in our data; the Le [2026] caveat applies only to interventions sharing a prompt channel. Library construction is therefore highly parallelisable.

Capability libraries are coarser than error-class libraries. Five named math classes (arithmetic, units, counting, list manipulation, date arithmetic) collapse under one Python interpreter; three code classes (SyntaxError, NameError, most TypeError) collapse under one execution-feedback loop; format violations and the structural component of “missing required element” (together more than 20% of ErrorAtlas) collapse under one constrained-decoder deployment. A team budgeting for ≈ 50 interventions per domain can expect to need fewer than 50 capability-axis interventions while covering ≈ 50 named clusters. The six axes of Appendix B are closer to the right unit of engineering accounting than the twelve clusters.

6 Discussion

By-construction elimination as a boundary case. A subset of the capability-elimination harvest deserves separate accounting. Seven of the 28 citations (Appendix B; constrained decoders [Suresh et al., 2025, Dong et al., 2025, Li et al., 2026, Zhang et al., 2023, OpenAI, 2024], proof kernels [Ren et al., 2025], static syntax checks) achieve residual error rate equal to zero *by mathematical construction*, not statistically. Constrained decoders set $P(\text{invalid token}) = 0$ at every generation step; the class of grammar-violating outputs is mathematically empty. In this regime the polylog bound of Proposition 2 is loose: covered clusters contribute exactly zero to residual error, not a small positive quantity. The result strengthens, rather than refutes, the framework. And Pattern A applies precisely to structural/verifiable classes (format, syntax, schema, invalid proofs), which sit at the head of the heavy-tailed cluster frequency distribution: the strongest mechanism lands exactly where the catalogue is densest.

Counter-evidence relocates, not dissolves. Five prominent steep-decay papers [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026, Karpinska et al., 2024] are routinely cited as exponential-compounding evidence. Every one decays over a variable distinct from raw token length: compositional graph size [Dziri et al., 2023], number of supporting facts [Kuratov et al., 2024], log-time horizon [Kwa et al., 2025], capacity threshold [Wan et al., 2026], evidence scope [Karpinska et al., 2024]. The apparent rapid decay is, in each case, in a quantity our framework already concentrates the action in (k_{hard} and $|C|$), not in raw sequence length: a relocation, not a dissolution. These regimes remain hard; the framework’s value is directing intervention toward capability provisioning along the actual decay axis rather than toward context-window expansion that does not help. Appendix C walks through each paper.

Limitations. Postulate 1 is empirical, not derived; a domain with genuinely Heaps-power-law mode discovery would invalidate the doubly-logarithmic special case while preserving the qualitative polylog conclusion. The coverage form of §3.3 is an empirical best-fit; readers should not interpret the numerical match against anchors as theoretical confirmation. Cluster-orthogonality is approximate; Le [2026]’s caveat on prompt-channel interference tightens the bound in shared-channel settings. Domain narrowness: Postulate 1’s empirical anchor rests on three taxonomies (ErrorAtlas, HumanEval, MWPEs), all published 2025–2026 and covering general/code/math; the framework is untested on agentic workflows, long-running scientific reasoning, and multi-turn tool use over millions of tokens, precisely the regimes where $|C|$ may grow faster than logarithmically. The $\sigma \approx 1.85$ estimate is a single-point calibration, not a discovery-curve fit; a subsample-vs-distinct-modes plot has not been published for any LLM error taxonomy at this writing. β is latent and we report no empirical range. Taxonomy granularity (the L2–L3 gap) is unmeasured. Patch-shift between deployments changes $A_D, \sigma_D, \beta_D, |C_D|$ simultaneously; libraries calibrated against one patch under-cover the next. Finally, the framework *relocates* the difficulty of long-context reliability rather than resolving

it: where k_{hard} grows with task length (adversarial compositional structure, multi-hop chains, long agent horizons), reliability remains hard. The contribution is to name the on-axis intervention, not to make those regimes easy. A falsifiability test the framework passes: DebugBench [Tian et al., 2024] explicitly catalogues which classes execution-feedback can and cannot repair, and the empirical split (works for syntax/reference; “unhelpful for logic errors”) is exactly what the framework’s selectivity claim predicts.

7 Conclusion

LLM reliability is often framed as an asymptotic scaling problem. Universal reliability is not a finite-library problem (Proposition 1): an unbounded domain that keeps producing intervention-distinguishable failures cannot be covered by any finite dictionary. The contribution of this paper is to show that, once a bounded deployment patch is fixed, reliability becomes a local engineering problem. Prior work in this line [Anonymous, 2025] argued that the standard asymptotic framing rests on a false uniformity assumption: errors concentrate at $\approx 5\text{--}10\%$ of tokens. This paper takes the next step: within that sparse set, errors are not only concentrated but also *repetitive*, clustering into a finite catalogue whose size grows logarithmically with observed failures under the empirical postulate of §3.2, or as a small power under the conservative Heaps alternative (Appendix A). The conditional consequence (Proposition 2) is that the intervention budget required to bound per-hard-token residual error scales polylogarithmically in sequence length within a fixed domain patch, and becomes a domain-constant once the patch ceiling $|C_D|$ is reached. Sequence-level reliability targets are strictly tighter. Available evidence is consistent with libraries on the order of tens of interventions covering the head of the per-hard-token failure distribution in many fixed domains. The exact number is patch-indexed and should be estimated from local discovery and rank-coverage curves, not assumed from cross-domain priors. This reframes the question from “can we bound the growth of n -token error?” to “have we catalogued enough failure modes *inside the deployment patch*?” The latter is finite and addressable once a bounded deployment patch has been fixed. Direct empirical measurement of the mode-rate constant σ on new domains (agentic, scientific, code-in-production) is the open empirical question this paper invites; future work flagged by Anonymous [2025, §6] on attention-mechanism bounds and stratified-manifold geometry would replace the empirical postulate with a derived result.

References

- Anonymous. Beyond exponential decay: Rethinking error accumulation in large language models. *arXiv preprint*, 2025. Author field anonymized for double-blind submission; arXiv ID public.
- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations (ICLR)*, 2024.
- Shir Ashury-Tahan, Yifan Mai, Elron Bandel, Michal Shmueli-Scheuer, and Leshem Choshen. ErrorMap and ErrorAtlas: Charting the failure landscape of large language models. *arXiv preprint*, 2026.
- Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. Why do multi-agent LLM systems fail? *arXiv preprint*, 2025. Introduces the MAST taxonomy (Multi-Agent System Failure Taxonomy).
- Wenhu Chen, Xueguang Ma, Xinyi Wang, and William W. Cohen. Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks. *Transactions on Machine Learning Research (TMLR)*, 2023.
- Kanzhi Cheng, Qiushi Sun, Yougang Chu, Fangzhi Xu, Yantao Li, Jianbing Zhang, and Zhiyong Wu. SeeClick: Harnessing GUI grounding for advanced visual GUI agents. In *Proceedings of ACL*, 2024.
- Caia Costello, Simon Guo, Anna Goldie, and Azalia Mirhoseini. Think, prune, train, improve: Scaling reasoning without scaling models. *arXiv preprint*, 2025.

467 Chengwei Dai et al. Capture the key in reasoning to enhance CoT distillation generalization. In
468 *Proceedings of ACL*, 2025. Earlier arXiv version titled “Beyond Imitation: Learning Key Reasoning
469 Steps from Dual Chain-of-Thoughts in Reasoning Distillation”.

470 Alex Dantart. Reliability by design: Quantifying and eliminating fabrication risk in LLMs. from
471 generative to consultative AI: A comparative analysis in the legal domain and lessons for high-
472 stakes knowledge bases. *arXiv preprint*, 2026.

473 Yixin Dong, Charlie F. Ruan, Yaxing Cai, Ruihang Lai, Ziyi Xu, Yilong Zhao, and Tianqi Chen.
474 XGrammar: Flexible and efficient structured generation engine for large language models. In
475 *Proceedings of MLSys*, 2025.

476 Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Peter West,
477 Chandra Bhagavatula, Ronan Le Bras, Jena D. Hwang, et al. Faith and fate: Limits of transformers
478 on compositionality. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

479 Lizhe Fang, Yifei Wang, Zhaoyang Liu, Chenyang Zhang, Stefanie Jegelka, Jianfeng Gao, Bolin
480 Ding, and Yisen Wang. What is wrong with perplexity for long-context language modeling? In
481 *International Conference on Learning Representations (ICLR)*, 2025.

482 Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and
483 Graham Neubig. PAL: Program-aided language models. In *Proceedings of ICML*, 2023a.

484 Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate
485 text with citations. In *Proceedings of EMNLP*, 2023b. Introduces the ALCE benchmark.

486 Alex J. Goodell, Simon N. Chu, Dara Rouholiman, and Larry F. Chu. Large language model
487 agents can use tools to perform clinical calculations. *npj Digital Medicine*, 8(1):163, 2025. doi:
488 10.1038/s41746-025-01475-8.

489 Boyu Gou, Ruohan Wang, Boyuan Zheng, Yanan Xie, Cheng Chang, Yiheng Shu, Huan Sun, and
490 Yu Su. Navigating the digital world as humans do: Universal visual grounding for GUI agents. In
491 *International Conference on Learning Representations (ICLR)*, 2025. ICLR 2025 Oral. Introduces
492 the UGround model.

493 Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen.
494 CRITIC: Large language models can self-correct with tool-interactive critiquing. In *International*
495 *Conference on Learning Representations (ICLR)*, 2024a.

496 Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Minlie Huang, Nan Duan, and
497 Weizhu Chen. ToRA: A tool-integrated reasoning agent for mathematical problem solving. In
498 *International Conference on Learning Representations (ICLR)*, 2024b.

499 Dadi Guo, Jiayu Liu, Zhiyuan Fan, Zhitao He, Haoran Li, Yuxin Li, Yumeng Wang, and Yi R. Fung.
500 Mathematical proof as a litmus test: Revealing failure modes of advanced large reasoning models.
501 *arXiv preprint*, 2025. Introduces the RFMDataset (Reveal Failure Modes).

502 Cheng-Ping Hsieh, Simeng Sun, Samuel Krizan, Shantanu Acharya, Dima Rekish, Fei Jia, Yang
503 Zhang, and Boris Ginsburg. RULER: What’s the real context size of your long-context language
504 models? In *Conference on Language Modeling (COLM)*, 2024.

505 Dong Huang, Qingwen Bu, Jie M. Zhang, Michael Luck, and Heming Cui. AgentCoder: Multi-agent-
506 based code generation with iterative testing and optimisation. *arXiv preprint*, 2023.

507 Nikhil Kandpal, Haikang Deng, Adam Roberts, Eric Wallace, and Colin Raffel. Large language
508 models struggle to learn long-tail knowledge. In *Proceedings of ICML*, 2023.

509 Batuhan K. Karaman, Ishmam Zabir, Alon Benhaim, Vishrav Chaudhary, Mert R. Sabuncu, and
510 Xia Song. POROver: Improving safety and reducing overrefusal in large language models with
511 overgeneration and preference optimization. *arXiv preprint*, 2024.

512 Marzena Karpinska, Katherine Thai, Kyle Lo, Tanya Goyal, and Mohit Iyyer. One thousand and one
513 pairs: A “novel” challenge for long-context language models. In *Proceedings of EMNLP*, 2024.

514 Yuri Kuratov, Aydar Bulatov, Pavel Anokhin, Ivan Rodkin, Dmitry Sorokin, Artyom Sorokin, and
515 Mikhail Burtsev. BABILong: Testing the limits of LLMs with long context reasoning-in-a-haystack.
516 In *NeurIPS Datasets and Benchmarks Track*, 2024.

517 Thomas Kwa et al. Measuring AI ability to complete long software tasks. In *Advances in Neural*
518 *Information Processing Systems (NeurIPS)*, 2025. METR technical report.

519 Yifan Le. Schema key wording as an instruction channel in structured generation under constrained
520 decoding. *arXiv preprint*, 2026.

521 Yann LeCun. Auto-regressive LLMs are doomed. Tweet, March 26, 2023. <https://x.com/ylecun/status/1640122342570336267>, 2023.

523 Linzhang Li, Yixin Dong, Guanjie Wang, Ziyi Xu, Alexander Jiang, and Tianqi Chen. XGrammar-2:
524 Efficient dynamic structured generation engine for agentic LLMs. *arXiv preprint*, 2026.

525 Xuyang Li and Anand D. Sarwate. Unraveling the localized latents: Learning stratified manifold
526 structures in LLM embedding space with sparse mixture-of-experts. *arXiv preprint*, 2025.

527 Yujia Li et al. Competition-level code generation with AlphaCode. *Science*, 378(6624):1092–1097,
528 2022. doi: 10.1126/science.abq1158.

529 Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan
530 Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International*
531 *Conference on Learning Representations (ICLR)*, 2024.

532 Fangyu Liu, Julian Martin Eisenschlos, Francesco Piccinno, Syrine Krichene, Chenxi Pang, Kenton
533 Lee, Mandar Joshi, Wenhua Chen, Nigel Collier, and Yasemin Altun. DePlot: One-shot visual
534 language reasoning by plot-to-table translation. In *Findings of ACL*, 2023.

535 Yadong Lu, Jianwei Yang, Yelong Shen, and Ahmed Awadallah. OmniParser for pure vision based
536 GUI agent. *arXiv preprint*, 2024.

537 Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi.
538 When not to trust language models: Investigating effectiveness of parametric and non-parametric
539 memories. In *Proceedings of ACL*, 2023. Introduces the PopQA benchmark.

540 Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. *Introduction to Information*
541 *Retrieval*. Cambridge University Press, 2008. Section on Heaps’ law: <https://nlp.stanford.edu/IR-book/html/htmledition/heaps-law-estimating-the-number-of-terms-1.html>.
542 <https://nlp.stanford.edu/IR-book/html/htmledition/heaps-law-estimating-the-number-of-terms-1.html>.
543 <https://nlp.stanford.edu/IR-book/html/htmledition/heaps-law-estimating-the-number-of-terms-1.html>.

544 Ayana Niwa and Hayate Iso. AmbigNLG: Addressing task ambiguity in instruction for NLG. In
545 *Proceedings of EMNLP*, 2024.

546 OpenAI. Introducing structured outputs in the API. OpenAI Engineering Blog. <https://openai.com/index/introducing-structured-outputs-in-the-api/>, 2024. Published August 6,
547 2024.
548 2024.

549 OpenAI. GPT-5 system card. <https://cdn.openai.com/gpt-5-system-card.pdf>, 2025. Pub-
550 lished August 7, 2025.

551 Jianhui Pang, Fanghua Ye, Derek F. Wong, Xun He, Wenxiang Chen, and Longyue Wang. Anchor-
552 based large language models. In *Findings of ACL*, 2024.

553 Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry
554 of large language models. In *Proceedings of ICML*, 2024.

555 Khush Patel, Siva Surendira, Jithin George, and Shreyas Kapale. The six sigma agent: Achieving
556 enterprise-grade reliability in LLM systems through consensus-driven decomposed execution.
557 *arXiv preprint*, 2026.

558 Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A. Smith, and Mike Lewis. Measuring
559 and narrowing the compositionality gap in language models. In *Findings of EMNLP*, 2023.

560 Z. Z. Ren et al. DeepSeek-Prover-V2: Advancing formal mathematical reasoning via reinforcement
561 learning for subgoal decomposition. *arXiv preprint*, 2025.

562 Rylan Schaeffer et al. How do large language monkeys get their power (laws)? In *International
563 Conference on Machine Learning (ICML)*, 2025.

564 Yu Shang, Yu Li, Keyu Zhao, Likai Ma, Jiahe Liu, Fengli Xu, and Yong Li. AgentSquare: Automatic
565 LLM agent search in modular design space. *arXiv preprint*, 2024.

566 Yuling Shi, Songsong Wang, Chengcheng Wan, Min Wang, and Xiaodong Gu. From code to
567 correctness: Closing the last mile of code generation with hierarchical debugging. *arXiv preprint*,
568 2024. Introduces the MGDebugger method.

569 Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu
570 Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural
571 Information Processing Systems*, volume 36, 2023.

572 Yuhong Sun, Zhangyue Yin, Xuanjing Huang, Xipeng Qiu, and Hui Zhao. Error classification of
573 large language models on math word problems: A dynamically adaptive framework. In *arXiv
574 preprint*, 2025. Introduces the MWPEs-300K dataset (304,865 error samples from 15 LLMs
575 across four MWP datasets).

576 Tarun Suresh et al. DINGO: Constrained inference for diffusion LLMs. *arXiv preprint*, 2025.

577 Manan Suri, Puneet Mathur, Nedim Lipka, Franck Dernoncourt, Ryan A. Rossi, and Dinesh Manocha.
578 Structured uncertainty guided clarification for LLM agents. *arXiv preprint*, 2025. Introduces the
579 SAGE-Agent method.

580 Runchu Tian, Yining Ye, Yujia Qin, Xin Cong, Yankai Lin, Zhiyuan Liu, and Maosong Sun. De-
581 bugBench: Evaluating debugging capability of large language models. In *Findings of ACL*,
582 2024.

583 Akihiko Wada et al. Retrieval-augmented generation elevates local LLM quality in radiology contrast
584 media consultation. *npj Digital Medicine*, 8(1):395, 2025. doi: 10.1038/s41746-025-01802-z.

585 Kai Wan et al. A Fano-style accuracy upper bound for LLM single-pass reasoning in multi-hop QA.
586 In *International Conference on Learning Representations (ICLR)*, 2026.

587 Benlu Wang et al. From scores to steps: Diagnosing and improving LLM performance in evidence-
588 based medical calculations. In *Proceedings of EMNLP*, 2025a. Introduces the MedRaC benchmark.

589 Darren Yow-Bang Wang, Zhengyuan Shen, Soumya Smruti Mishra, Zhichao Xu, Yifei Teng, and
590 Haibo Ding. SLOT: Structuring the output of large language models. *arXiv preprint*, 2025b. SLOT
591 = Structured LLM Output Transformer.

592 Minzheng Wang et al. Leave no document behind: Benchmarking long-context LLMs with extended
593 multi-doc QA. In *Proceedings of EMNLP*, 2024a.

594 Peiyi Wang, Lei Li, Zhihong Shao, R. X. Xu, Damai Dai, Yifei Li, Deli Chen, Y. Wu, and Zhifang
595 Sui. Math-Shepherd: Verify and reinforce LLMs step-by-step without human annotations. In
596 *Proceedings of ACL*, 2024b.

597 Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdh-
598 ury, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models.
599 In *International Conference on Learning Representations (ICLR)*, 2023.

600 Yubo Wang et al. MMLU-Pro: A more robust and challenging multi-task language understanding
601 benchmark. In *NeurIPS Datasets and Benchmarks Track (Spotlight)*, 2024c.

602 Hao Wen et al. Fixing function-level code generation errors for foundation large language models.
603 *arXiv preprint*, 2024. Introduces the LlmFix method. Reports AssertionError 63.64% + NameError
604 22.71% = 86.35% of HumanEval failures across 14 LLMs.

605 Michael C. Wood and Adam A. Forbes. 100% elimination of hallucinations on RAGTruth for GPT-4
606 and GPT-3.5 Turbo. *arXiv preprint*, 2024. Introduces the Acurai method.

- 607 Tianbao Xie et al. Scaling computer-use grounding via user interface decomposition and synthesis.
608 In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. NeurIPS 2025 Spotlight.
609 Introduces the Jedi grounding dataset and OSWorld-G benchmark.
- 610 Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -bench: A benchmark for
611 tool-agent-user interaction in real-world domains. *arXiv preprint*, 2024.
- 612 Cyril Zakka, Rohan Shad, Akash Chaurasia, Alex R. Dalal, Jennifer L. Kim, Michael Moor, Robyn
613 Fong, Curran Phillips, Kevin Alexander, Euan Ashley, Jack Boyd, Kathleen Boyd, Karen Hirsch,
614 Curtis Langlotz, Rita Lee, Joanna Melia, Joanna Nelson, Karim Sallam, Stacey Tullis, Melissa Ann
615 Vogelsong, John Patrick Cunningham, and William Hiesinger. Almanac — retrieval-augmented
616 language models for clinical medicine. *NEJM AI*, 1(2):AIoa2300068, 2024. doi: 10.1056/
617 AIoa2300068.
- 618 Kexun Zhang et al. Don’t fine-tune, decode: Syntax error-free tool use via constrained decoding.
619 *arXiv preprint*, 2023. Introduces the ToolDec method.
- 620 Meiru Zhang, Zaiqiao Meng, and Nigel Collier. Failure modes in multi-hop QA: The weakest link
621 effect and the recognition bottleneck. *arXiv preprint*, 2026.
- 622 Yang Zhou, Hongyi Liu, Zhuoyan Chen, Yuandong Tian, and Beidi Chen. GSM-Infinite: How do
623 your LLMs behave over infinitely increasing context length and reasoning complexity? *arXiv*
624 *preprint*, 2025.

625 A Heaps Power-Law Variant and Cluster-Count Sensitivity

626 A reader who prefers to derive cluster-count growth from standard Heaps’ law rather than from
627 Postulate 1 obtains a qualitatively similar result. Let $|C|(k_{\text{hard}}) \approx K \cdot k_{\text{hard}}^b$ with $b \in (0, 1)$. Canonical
628 fits give $b \approx 0.5$ for natural-language vocabularies; failure-mode taxonomies plateau much more
629 sharply (ErrorAtlas stabilises at $|C| = 17$ across 10^4 failures), implying a smaller effective $b \approx 0.2$
630 in our setting. Composing with $k = \Theta(\log n)$, we get $|C| = O((\log n)^b)$, and Proposition 2 becomes

$$m = O\left((\log n)^{b \cdot (1 - \varepsilon / e_{\text{hard}})}\right).$$

631 This is still polylogarithmic in n for any $b \in (0, 1)$ and any $\varepsilon < e_{\text{hard}}$. The paper’s qualitative claim
632 survives either choice of cluster-count law.

633 **Symbolic-form sensitivity across candidate laws.** The polylog conclusion depends on which
634 cluster-count law one accepts; available evidence is consistent with multiple candidates because no
635 subsample-discovery curve has been published for any LLM failure-mode taxonomy at this writing.
636 We therefore report symbolic rates rather than fitted constants. With $h(n) = \beta k(n)$:

- 637 • **Logarithmic:** $|C_{\text{active}, D}(n)| = O(\log h(n))$. If $k(n) = \Theta(\log n)$, then $m =$
638 $O((\log \log n)^{1 - \varepsilon / e_{\text{hard}}})$.
- 639 • **Heaps:** $|C_{\text{active}, D}(n)| = O(h(n)^b)$ with $b \in (0, 1)$. If $k(n) = \Theta(\log n)$, then $m =$
640 $O((\log n)^{b(1 - \varepsilon / e_{\text{hard}})})$.
- 641 • **Saturating:** $|C_{\text{active}, D}(n)| \leq |C_D|$. Then $m = O(|C_D|^{1 - \varepsilon / e_{\text{hard}}})$, constant in n once the
642 patch ceiling is reached.

643 The qualitative conclusion is robust **under the $k(n) = \Theta(\log n)$ regime**: m grows more slowly than
644 any positive power of n under every candidate, and the directional claim (“a small library covers the
645 head of the failure distribution in the per-hard-token regime”) survives. Only the exponent shifts:
646 doubly-logarithmic under logarithmic discovery, $(\log n)^b$ with small b under Heaps, constant in the
647 cap regime. The doubly-logarithmic rate is the optimistic special case. If $k(n)$ grows as a positive
648 power of n , the Heaps variant inherits that power and the polylog-in- n language fails along that axis;
649 the framework’s intervention prescription still applies, but its asymptotic-rate framing does not.

650 Numerical constants require a measured discovery curve $C_{\text{seen}, D}(T)$ or $C_{\text{active}, D}(n)$. Existing tax-
651 onomies provide endpoint category counts at a single corpus scale (ErrorAtlas at $|C| = 17$ for

652 $\approx 10^4$ failures), not discovery curves. The subsample-discovery measurement remains the explicit
 653 empirical test that would either tighten the postulate or fall back to the Heaps variant. Until that
 654 measurement exists, the framework’s headline rate should be read as *polylogarithmic in the pre-cap*
 655 *regime, domain-constant in the cap regime*, with the specific exponent flagged as a falsifiability test
 656 rather than a fitted prediction.

657 B Full Failure-Mode Taxonomy and the Capability-Elimination Harvest

658 The intervention literature provides at least one targeted countermeasure for each named cluster of §4.
 659 Two organising granularities exist: at the capability level (six axes) the cluster-orthogonality property
 660 is most clearly visible; at the error-class level (twelve named clusters) the evidence aligns with the
 661 taxonomies of §4 but is finer-grained than the underlying capability mechanisms.

662 **Six capability axes and three structural patterns.** A dedicated harvest yields **28 quantitatively-**
 663 **anchored citations** across **six independent capability axes**: Arithmetic (Python/symbolic execution),
 664 Code Execution (REPL/sandbox feedback), Format/Structure (constrained decoding, FSMs, grammar
 665 engines), Perception/Grounding (visual grounding for GUI, charts, tables), Knowledge/RAG (dense
 666 retrieval and citation grounding), Verification (proof checkers, learned verifiers, classifier rerouting,
 667 process supervision). Each axis is independently confirmed by between three and nine citations. We
 668 stratify the 28 by kind of evidence into three patterns:

669 **Pattern A: hard guarantees (by-construction).** Seven citations achieve residual error rate equal
 670 to zero *by construction*, restricted strictly to structural/verifiable classes: constrained decoders set
 671 $P(\text{invalid token}) = 0$ at every step [Suresh et al., 2025, Zhang et al., 2023, Dong et al., 2025,
 672 Li et al., 2026, OpenAI, 2024]; static syntax checks reject programs with a `SyntaxError` before
 673 execution [Wen et al., 2024]; proof kernels reject any output failing type-checking [Ren et al., 2025].
 674 The class of grammar-violating outputs is mathematically empty under these mechanisms.

675 **Pattern B: strong empirical reductions with class-shift signature.** Roughly fourteen citations
 676 report empirical reductions of 80–100% in a named error class, with the post-intervention failure
 677 log dominated by structurally different residual classes. Program-of-Thoughts on GSM8K [Chen
 678 et al., 2023]: calculation errors drop from 30% of failures to 0%, residuals are 62% reasoning + 36%
 679 misunderstanding. OpenMedCalc [Goodell et al., 2025]: “only interpretation errors were identified.”
 680 Acurai [Wood and Forbes, 2024]: 100% hallucination elimination on RAGTruth (95% CI 91–100%),
 681 strong empirical rather than by-construction.

682 **Pattern C: moderate reductions (60–80%) with residuals shifting outside the target class.**
 683 Legal RAG [Dantart, 2026]: fabricated citations $> 30\% \rightarrow < 0.2\%$. GPT-5 SimpleQA with web
 684 access [OpenAI, 2025]: $47\% \rightarrow 9.6\%$ (inter-condition, not within-condition). CRITIC [Gou et al.,
 685 2024a]: toxic generation -79.2% .

686 The 28 citations, organised by axis and pattern.

Axis	Citation & targeted class	Pre → Post	Pattern
Verification	Ren et al. [2025]: Invalid Lean proof	any → 0% by constr.	A
Format	OpenAI [2024]: JSON schema violation	>60% → 0% by constr.	A
Format	Zhang et al. [2023]: Tool-call syntax	21–100% → 0%	A
Format	Suresh et al. [2025]: JSON parse failure	13–82% → 0%	A
Format	Dong et al. [2025]: Multi-format errors	20–38% → 0%	A
Format	Li et al. [2026]: Malformed tool calls	33–78% → 0%	A
Arithmetic	Chen et al. [2023]: Calculation on GSM8K	30% → 0% of failures	B
Arithmetic	Goodell et al. [2025]: Clinical arithmetic	“only interpretation errors”	B
Knowledge/RAG	Wada et al. [2025]: RAG hallucinations	8% → 0% ($p = 0.012$)	B
Knowledge/RAG	Wood and Forbes [2024]: Context-conflict hallucinations	100% → 0% on subset	B
Knowledge/RAG	Dantart [2026]: Fabricated legal citations	>30% → <0.2%	C
Code Exec	Wen et al. [2024]: <code>SyntaxError</code> (HumanEval)	5.76% → 0.01%	A
Code Exec	Li et al. [2022]: False-positive submissions	62% → 4%	B
Code Exec	Shi et al. [2024]: 5 code-bug classes	100% repair (5/6 classes)	B
Knowledge/RAG	Gao et al. [2023b]: Citation grounding (ELI5)	≈50% w/o complete support	C
Knowledge/RAG	OpenAI [2025]: Factual errors	47% → 9.6% w/web	C
Knowledge/RAG	Zakka et al. [2024]: Clinical citation errors	44% → 9%	C
Arithmetic	Wang et al. [2025a]: Arithmetic in medical reasoning	426 → 74 errors	B
Code Exec	Wen et al. [2024]: <code>NameError</code> repair	22.7% → 2.3%	C
Perception	Gou et al. [2025]: GUI grounding errors	16.2% → 73.3% acc.	B
Perception	Xie et al. [2025]: GUI task failures	5% → 27% SR ($5.4\times$)	B
Perception	Cheng et al. [2024]: Element mislocation	5.2% → 53.4% ($10.3\times$)	B
Perception	Liu et al. [2023]: Chart-reading errors	38.2% → 67.6% (+29.4 pp)	B
Verification	Gou et al. [2024a]: Toxic generation	−79.2%	C
Verification	Wang et al. [2024b]: Reasoning step errors	28.6% → 43.5% w/rerank	B
Knowledge/RAG	Asai et al. [2024]: Unsupported facts	55.5% → 22% error	B
Perception	Lu et al. [2024]: Icon/element errors	70.5% → 93.8% (79% red.)	B
Knowledge/RAG	Mallen et al. [2023]: Long-tail entity errors	≈80% → ≈50% failure	B

Capabilities are coarser than error classes. Several entries in the harvest reveal “two-for-one” reductions where one capability addresses multiple named clusters: a single Python interpreter removes the execution-error component of arithmetic, unit conversion, simple counting, list manipulation, and date arithmetic; constrained decoding eliminates by construction both format violations and the structural component of “missing required element”; code execution feedback strongly reduces `SyntaxError`, `NameError`, and most `TypeError` together; RAG strongly reduces factual hallucinations, fabricated citations, and outdated information jointly. The practical capability library required is therefore smaller than the count of named error categories.

The twelve named clusters.

Cluster	Axis	Intervention	Before → After	Patt.
A. Arithmetic	Arithmetic	PAL Python [Gao et al., 2023a]	20.1% → 61.5%	B
B. Unit conversion	Arithmetic	Folded into A (Python)	n/a	n/a
C. Counting	Arithmetic	Explicit counter (gap)	n/a	gap
D. Format/schema	Format	DINGO/SLOT [Suresh et al., 2025, Wang et al., 2025b]	up to +68 pp; 99.5% acc.	A
E. Code logic/type	Code Exec	Reflexion/AgentCoder [Shinn et al., 2023, Huang et al., 2023]	80% → 96.3% pass@1	A/B
F. Multi-hop drift	Knowledge	Entity-grounded rewriter	≈5–10 pp	C
G. Reasoning step	Verification	Math-Shepherd [Wang et al., 2024b]	28.6% → 43.5%	B
H. Spec misinterpret.	Verification	Clarification [Niwa and Iso, 2024]	ROUGE-L +15	C
I.a Struct. missing	Format	Subsumed by D (constr. decode)	n/a	A
I.b Semantic missing	Verification	Subsumed by G/H	n/a	C
J. Hallucination	Knowledge	RAG [Wood and Forbes, 2024]	non-zero → 0%	B/C
K. Refusal	Verification	POROver [Karaman et al., 2024]	57.6% → 82.1%	B
L. Tool/API	Format+Verif.	SAGE-Agent [Suri et al., 2025]	36.5% → 65.2%	B

Eight of twelve categories have a strong citation with double-digit percentage-point improvement; three (B, C, I) lack a clean single-cluster ablation; one (F) has consistent medium-strength evidence. Category I (missing required elements) splits mechanistically into I.a (structural, absorbed by D’s constrained decoders) and I.b (semantic, absorbed by G/H). Categories B (units) and C (counting) remain technical gaps: B is naturally folded into A; C is the smallest residual gap.

Additivity and its limits. Patel et al. [2026] demonstrate that stacking interventions targeting orthogonal failure modes can produce large compound reliability gains: their parallel-consensus framework yields a $14,700\times$ improvement over single-pass baseline, evidence for compound gains from decomposition plus consensus aggregation under their specific parallel-voting regime rather than a direct demonstration that heterogeneous cluster-targeted interventions compose additively without voting. Shang et al. [2024] similarly show supra-additive gains when modules target distinct failure modes. Le [2026] reports that schema-level and prompt-level instructions interact *non-additively* when sharing a prompt channel: interventions on orthogonal processing layers (decoding constraint vs. retrieval vs. training-signal vs. inference-time tool call) compose near-additively, while those sharing a channel may interfere.

The irreducible-semantic residual. Of the 17 ErrorAtlas categories, 13 are addressed under Patterns A, B, or C by one of the six capability axes; four are not: residuals of inappropriate refusal beyond preference optimisation, specification misinterpretation not closed by clarification, problem-decomposition reasoning bottlenecks, and a “user wanted something different” semantic remainder. These are classes where the failure is in choosing what to do, not executing it. Proposition 2’s prediction of $O(\log)$ rather than $O(0)$ residual reliability reflects exactly this irreducible core. The framework does not claim 100% coverage of all failures; it claims polylog-bounded *capability-eliminable* failures, with the named semantic residual as the floor.

C Counter-Evidence Re-Audits

Five prominent papers are routinely cited as evidence that LLM reliability decays steeply with length [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026, Karpinska et al., 2024]. A careful re-reading shows that *every one* of these papers decays over a variable distinct from raw token length n . Identifying the decay axis is not the same as dissolving the practical concern: where compositional graph size, fact count, or evidence scope grow with problem length, the framework predicts steep failure curves. The contribution is to identify which interventions help (capability provisioning along the actual decay axis) and which do not.

Dziri et al. [2023], Faith and Fate. GPT-4 multi-digit multiplication accuracy drops from 59% (3-digit) to 4% (4-digit) zero-shot, with the authors theorising “probability of incorrect predictions

converges exponentially to ≈ 1 for abstract compositional tasks.” The decay variable is *compositional graph size* N , not raw token length: the 3×3 graph has on the order of d^2 partial products plus carries, and multi-digit multiplication is engineered so that $k_{\text{hard}} \approx N$, with every node in the computation graph a hard decision. For natural-language tasks where $k_{\text{hard}} \ll n$, Dziri’s regime is the *boundary case* in which our framework reduces to their result. A clean $(1 - \varepsilon)^N$ exponential cannot simultaneously reproduce 59% at 3-digit and 4% at 4-digit for any single per-node ε : the observed drop is locally steeper than per-node iid exponential, consistent with a finite catalogue of failure modes exhausting as N grows. *Where the framework concedes*: in adversarial compositional tasks engineered so $k_{\text{hard}} \approx N$, the framework reduces to Dziri’s regime and does not relieve it; the relocation is informational, not magical.

Kuratov et al. [2024]. The abstract reports “performance declines sharply with increased reasoning complexity” and models “effectively utilise only 10–20% of the context.” Read in detail, BABILong varies two axes: context length n (0K to 10M tokens) and number of supporting facts k (QA1 = 1 fact to QA3 = 3 facts). The sharp decline is in k , not n . For QA1 (single-fact), most models “perform well up to 4,000 tokens”: a plateau, not exponential decay. The famous RAG-flat-across-length result (60% on single-fact QA *independent of context length*) is the cleanest possible demonstration that when relevant evidence is in window, length does not matter. Recurrent Memory Transformers maintaining performance to 50M tokens further confirms effective k is determined by architecture, not raw n . The framework’s concession is narrow but real: for multi-hop tasks whose required fact count grows with task complexity, BABILong’s sharp k -axis decay is exactly what the framework predicts happens, not a counter-example.

Kwa et al. [2025] (METR). Per-model success fits a logistic in $\log(\text{human-task-duration})$: $S(t) = \sigma(\beta(\log t - \log h))$. Logistic-in-log-length is *mathematically sublinear* in length itself. The 80%-horizon being $4\text{--}6\times$ shorter than the 50%-horizon is a steep within-model cliff but compatible by construction with the polylog result: a cliff at a specific capacity threshold is a manifold-transition signature. The famous exponential (capability doubling every seven months) is an *inter-model* claim about how the horizon h moves across generations, orthogonal to within-model decay shape. An honest qualifier: the within-model logistic-in-log cliff is steep on a practical scale. Calling it “sublinear in length” is technically correct but engineering-useful only for systems designed to operate well below the 50% horizon.

Wan et al. [2026], Fano-style upper bound. This paper theorises super-linear information-demand growth and identifies an “accuracy cliff” at capacity overflow: “when the task’s information demand surpasses the model’s output capacity, performance does not degrade gracefully but instead collapses sharply.” The behavioural prediction is a *threshold*, not smooth $(1 - \varepsilon)^n$. A cliff at a specific capacity threshold is exactly the manifold-transition behaviour our framework predicts at the boundary between covered and uncovered modes; Wan et al.’s mechanism (capacity overflow) is one specific cause, complementary to ours. *Where the framework concedes*: Wan documents a mechanism by which $|C|$ effectively exceeds the model’s coverage in a single forward pass; Pattern A interventions (constrained decoding, formal verification) are the kind of response the framework expects but does not automatically supply.

Karpinska et al. [2024]. GPT-4o achieves 55.8% pair accuracy across 1,001 minimally-different true/false claim pairs about long fictional books (mean length 127K tokens). The paper reports performance by *evidence scope*, not by context length: 59.8% on sentence-level retrieval, 47.6% on passage-level, 41.6% on global reasoning. No per-context-length curve is reported. The decay axis is the number of evidence pieces that must be integrated, a k -axis quantity rather than an n -axis one. NoCha is therefore not counter-evidence to a sublinear-in- n claim. Evidence-scope decay is exactly the k -axis observation the framework predicts cannot be addressed by scaling raw context length; it requires retrieval or process supervision along the actual decay axis. The relocation is informational, not magical.

Unifying observation. Every counter-paper decays over a variable that is not n : compositional graph size, fact count, log-time horizon, capacity threshold, or evidence scope. The apparent rapid decay is, in each case, in a quantity our framework already concentrates the action in (k_{hard} and $|C|$), not in raw sequence length. This is a relocation, not a dissolution. Where k_{hard} grows with task length (adversarial compositional structure, multi-hop fact chains, long horizons that force more

786 decisions), reliability remains hard; the framework's value is directing intervention toward capability
787 provisioning along the actual decay axis rather than toward context-window or compute-budget
788 expansion that does not help.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and §1 state five contributions: (i) the three-layer framework partitioning key tokens into easy/hard subsets with a bounded mode catalogue (§3); (ii) the empirical postulate of logarithmic mode discovery anchored against three taxonomies with $\sigma \in [0.87, 1.85]$ (§3.2); (iii) Proposition 1 on the polylog intervention budget, with explicit per-hard-token vs. sequence-level distinction; (iv) evidence synthesis for clustering, cluster-selective interventions, and sublinear length scaling (§4); (v) the counter-evidence re-audit (Appendix C). Each claim is developed in the cited section. The paper makes no original empirical claims and is scoped as a framework-plus-synthesis contribution; limitations are explicit (§6).

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: §6 enumerates eight specific limitations: (i) Postulate 1 is empirical, not derived—a genuinely Heaps-power-law domain would invalidate the doubly-logarithmic special case (while preserving the qualitative polylog conclusion of Appendix A); (ii) the coverage form is an empirical best-fit, not theoretically derived; (iii) cluster-orthogonality is approximate, with Le [2026]’s prompt-channel-interference caveat noted; (iv) domain narrowness—the empirical anchor rests on three taxonomies covering general/code/math, with agentic and scientific regimes untested; (v) the $\sigma \approx 1.85$ estimate is single-point calibration, not a discovery-curve fit; (vi) β is latent and we report no empirical range; (vii) the L2–L3 taxonomy-granularity gap is unmeasured; (viii) patch-shift between deployments changes $A_D, \sigma_D, \beta_D, |C_D|$ simultaneously. The closing “goalpost-relocation caveat” makes explicit that the framework *relocates* the difficulty rather than resolving it: where k_{hard} grows with task length, reliability remains hard, and the framework’s value is in naming the on-axis intervention.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.

- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper develops a descriptive/conditional framework rather than formal theorems. Proposition 1 is engineering math whose conditional nature is stated explicitly in the surrounding “Conditionality” paragraph and in its statement: it holds conditional on (i) the log-coverage form of §3.3 on its declared domain $m \geq 2$, $|C_{\text{eff}}| \geq 2$; (ii) the residual target $\varepsilon \in (0, e_{\text{hard}})$; (iii) Assumption 2 (per-sequence active-mode exposure) for any sequence-length rate claim. The two-line proof immediately following the proposition statement traces all algebraic steps. Postulate 1 is explicitly labelled an empirical postulate, not a theorem (this is the load-bearing math-claims discipline of the paper). We label the budget statement a “Proposition” rather than a “Theorem” to keep its conditional nature visible.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The paper is a framework-plus-synthesis contribution, not an empirical paper; it runs no new experiments and trains no new models. Every numerical value cited (§4, Appendix B) is pinned to a specific published source with the relevant benchmark, model, and result identified. The σ calibration in §3.2 cites ErrorAtlas, HumanEval, and MWPEs with the endpoint corpus sizes and category counts used. The 28 capability-elimination citations in Appendix B report pre/post percentages from the original sources. A reader following each citation can verify or reproduce the underlying empirical result by consulting the cited work.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [N/A]

Justification: The paper releases no new datasets, models, or code. It is a framework-and-synthesis paper; all underlying empirical results are owned by their respective authors and accessed through the cited references.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.

950 • At submission time, to preserve anonymity, the authors should release anonymized
951 versions (if applicable).

952 • Providing as much information as possible in supplemental material (appended to the
953 paper) is recommended, but including URLs to data and code is permitted.

954 **6. Experimental setting/details**

955 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
956 rameters, how they were chosen, type of optimizer) necessary to understand the results?

957 Answer: [Yes]

958 Justification: §4 specifies which benchmarks supply evidence for each load-bearing claim:
959 ErrorAtlas (83 models $\times$ 35 datasets) and MWPEs (304,865 errors, 15 models) for cluster-
960 ing; the 28 capability-elimination interventions (Appendix B) for cluster-selectivity, with
961 per-row pre/post percentages and the model/benchmark named; Loong, GSM- ∞ , RULER,
962 Anchor LLMs, and Press et al. for sublinear length scaling, each with the specific length
963 range, model, and metric identified. The σ calibration in §3.2 cites three taxonomies with
964 k_{hard} and $|C|$ endpoint values used in the formula. No original training or hyperparameter
965 selection is performed by this paper.

966 Guidelines:

967 • The answer [N/A] means that the paper does not include experiments.

968 • The experimental setting should be presented in the core of the paper to a level of detail
969 that is necessary to appreciate the results and make sense of them.

970 • The full details can be provided either with the code, in appendix, or as supplemental
971 material.

972 **7. Experiment statistical significance**

973 Question: Does the paper report error bars suitably and correctly defined or other appropriate
974 information about the statistical significance of the experiments?

975 Answer: [N/A]

976 Justification: The paper reports no original experimental results, hence no error bars. Where
977 numerical results are cited from the literature (e.g., the +41.4-pp PAL gain, the 100%
978 Acurai elimination on RAGTruth with 95% CI 91–100%, the 32.9% Loong endpoint), the
979 originating work is cited with its reported significance treatment.

980 Guidelines:

981 • The answer [N/A] means that the paper does not include experiments.

982 • The authors should answer [Yes] if the results are accompanied by error bars, confidence
983 intervals, or statistical significance tests, at least for the experiments that support the
984 main claims of the paper.

985 • The factors of variability that the error bars are capturing should be clearly stated (for
986 example, train/test split, initialization, random drawing of some parameter, or overall
987 run with given experimental conditions).

988 • The method for calculating the error bars should be explained (closed form formula,
989 call to a library function, bootstrap, etc.)

990 • The assumptions made should be given (e.g., Normally distributed errors).

991 • It should be clear whether the error bar is the standard deviation or the standard error
992 of the mean.

993 • It is OK to report 1-sigma error bars, but one should state it. The authors should
994 preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis
995 of Normality of errors is not verified.

996 • For asymmetric distributions, the authors should be careful not to show in tables or
997 figures symmetric error bars that would yield results that are out of range (e.g., negative
998 error rates).

999 • If error bars are reported in tables or plots, the authors should explain in the text how
1000 they were calculated and reference the corresponding figures or tables in the text.

1001 **8. Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [N/A]

Justification: The paper performs no original computational experiments. Compute requirements of the cited systems are documented in those works.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: We have reviewed the NeurIPS Code of Ethics. The work is theoretical/synthesis with no human subjects, no scraped or sensitive data, no model training, and no deployed system. There are no fairness, privacy, or security risks specific to the paper.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: §5 discusses positive implications: targeted intervention-library design, SLA alignment between per-hard-token and sequence-level reliability targets, and parallelisable library construction. The negative-impact direction is structurally limited because the paper releases no model or dataset; the closing “goalpost-relocation caveat” (§6) explicitly warns that the framework *relocates* the difficulty of long-context reliability and should not be misread as a license to deploy LLMs in higher-stakes settings than the cited evidence justifies.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate

1054 to point out that an improvement in the quality of generative models could be used to
 1055 generate Deepfakes for disinformation. On the other hand, it is not needed to point out
 1056 that a generic algorithm for optimizing neural networks could enable people to train
 1057 models that generate Deepfakes faster.

- 1058 • The authors should consider possible harms that could arise when the technology is
 1059 being used as intended and functioning correctly, harms that could arise when the
 1060 technology is being used as intended but gives incorrect results, and harms following
 1061 from (intentional or unintentional) misuse of the technology.
- 1062 • If there are negative societal impacts, the authors could also discuss possible mitigation
 1063 strategies (e.g., gated release of models, providing defenses in addition to attacks,
 1064 mechanisms for monitoring misuse, mechanisms to monitor how a system learns from
 1065 feedback over time, improving the efficiency and accessibility of ML).

1066 11. Safeguards

1067 Question: Does the paper describe safeguards that have been put in place for responsible
 1068 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
 1069 image generators, or scraped datasets)?

1070 Answer: [N/A]

1071 Justification: No new pretrained models, generative artifacts, or scraped datasets are released.
 1072 The contribution is a theoretical framework and a synthesis of cited work.

1073 Guidelines:

- 1074 • The answer [N/A] means that the paper poses no such risks.
- 1075 • Released models that have a high risk for misuse or dual-use should be released with
 1076 necessary safeguards to allow for controlled use of the model, for example by requiring
 1077 that users adhere to usage guidelines or restrictions to access the model or implementing
 1078 safety filters.
- 1079 • Datasets that have been scraped from the Internet could pose safety risks. The authors
 1080 should describe how they avoided releasing unsafe images.
- 1081 • We recognize that providing effective safeguards is challenging, and many papers do
 1082 not require this, but we encourage authors to take this into account and make a best
 1083 faith effort.

1084 12. Licenses for existing assets

1085 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
 1086 the paper, properly credited and are the license and terms of use explicitly mentioned and
 1087 properly respected?

1088 Answer: [N/A]

1089 Justification: The paper uses no datasets, code, or models—it cites published work but
 1090 does not redistribute any artifact requiring license disclosure. Citations follow NeurIPS
 1091 conventions.

1092 Guidelines:

- 1093 • The answer [N/A] means that the paper does not use existing assets.
- 1094 • The authors should cite the original paper that produced the code package or dataset.
- 1095 • The authors should state which version of the asset is used and, if possible, include a
 1096 URL.
- 1097 • The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- 1098 • For scraped data from a particular source (e.g., website), the copyright and terms of
 1099 service of that source should be provided.
- 1100 • If assets are released, the license, copyright information, and terms of use in the
 1101 package should be provided. For popular datasets, paperswithcode.com/datasets
 1102 has curated licenses for some datasets. Their licensing guide can help determine the
 1103 license of a dataset.
- 1104 • For existing datasets that are re-packaged, both the original license and the license of
 1105 the derived asset (if it has changed) should be provided.

- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: No new assets are released by this paper.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing or research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human subjects research is conducted, so no IRB approval is required.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

1157 Question: Does the paper describe the usage of LLMs if it is an important, original, or
1158 non-standard component of the core methods in this research? Note that if the LLM is used
1159 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
1160 scientific rigor, or originality of the research, declaration is not required.

1161 Answer: [N/A]

1162 Justification: LLMs were used only for writing assistance (drafting, polishing, copy-editing)
1163 and not as part of the research methodology. The framework, the synthesis, and the choice
1164 of cited evidence are the authors'; LLMs were not involved in selecting evidence, deriving
1165 the framework, or shaping the argument. Per the question's stated guidance, no declaration
1166 is required for editing-only usage; we answer [N/A] accordingly.

1167 Guidelines:

- 1168 • The answer [N/A] means that the core method development in this research does not
1169 involve LLMs as any important, original, or non-standard components.
- 1170 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
1171 be described.